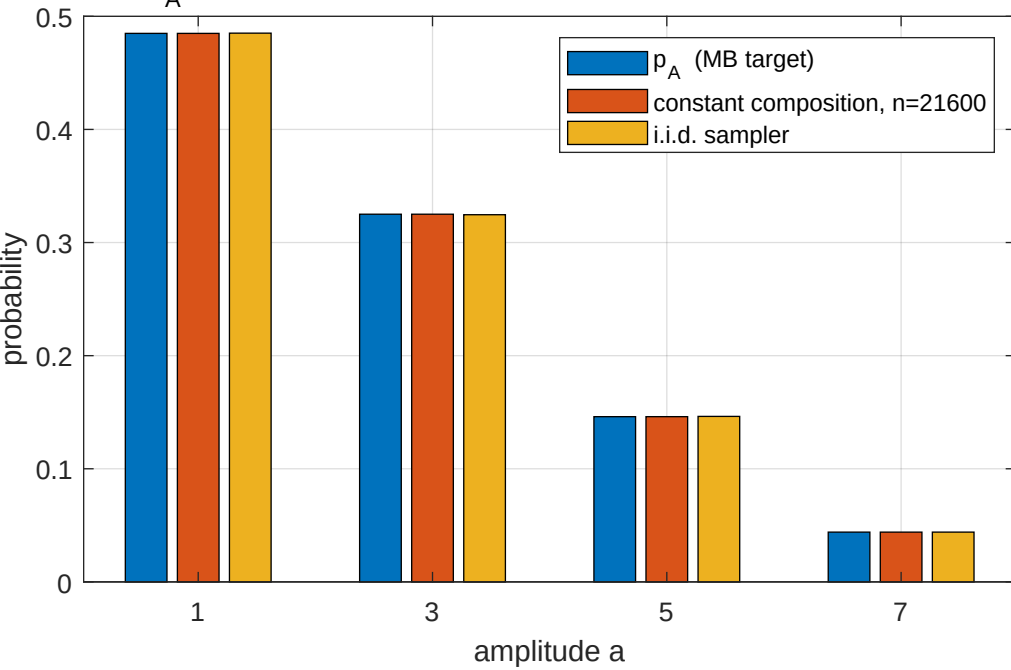


Shaped source: target vs transmitted

(a) amplitude distribution, $\nu=0.05$

$D(\cdot \| p_A)$: const.comp. $2.0\text{e-}09$ i.i.d. $8.0\text{e-}07$ VD-type($n=540$) $5.9\text{e-}06$



(b) $P(x_I, x_Q)$

E_s 18.4 vs 42.0 uniform: 3.58 dB less energy at fixed $d_{\min}$ (rate 1.637 vs 2.000 bit/amp)

